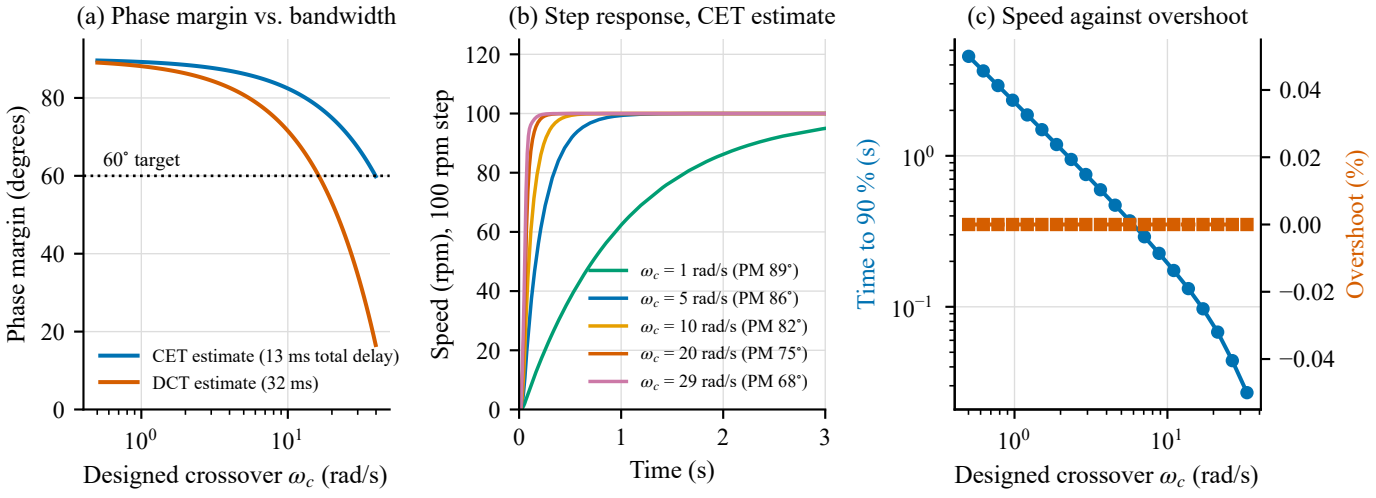

(a) Phase margin vs. bandwidth
Phase margin (degrees)
80
60
40
20
0
60° target
CET estimate (13 ms total delay)
DCT estimate (32 ms)
10^0
10^1
Designed crossover ω_c (rad/s)
(b) Step response, CET estimate
Speed (rpm), 100 rpm step
120
100
80
60
40
20
0
ω_c = 1 rad/s (PM 89°)
ω_c = 5 rad/s (PM 86°)
ω_c = 10 rad/s (PM 82°)
ω_c = 20 rad/s (PM 75°)
ω_c = 29 rad/s (PM 68°)
0
1
2
3
Time (s)
(c) Speed against overshoot
Time to 90 % (s)
10^0
10^{-1}
Overshoot (%)
0.04
0.02
0.00
−0.02
−0.04
10^0
10^1
Designed crossover ω_c (rad/s)